

A Case Study of Parkinson's Disease Using Machine Learning for Early Diagnosis and Progression Prediction of Neurodegenerative Diseases

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Abstract: Early diagnosis of Parkinson's Disease (PD) is crucial for effective clinical invention, yet the optimal combination of biomarkers and predictive model remains a subject of ongoing research. This study proposes a robust machine learning framework utilizing multimodal clinical data from the Parkinson's Progression Markers Initiative (PPMI) database. Our approaches integrate rigorous data preprocessing, SMOTE-based class balancing, and features optimization. We evaluated multiple supervised learning algorithms, specifically Random Forest (RF) and Support Vector Machine (SVM). While initial models showed signs of overfitting, refined testing after feature decorrelation demonstrated that the Random Forest model achieved superior performance with an accuracy of 0.96, alongside high precision and recall. These results underscore the efficacy of interpretable ML frameworks in identifying high-risk prodromal stages and supporting clinical decision-making.

Keywords: Parkinson disease, Neurogenerative disease, AI, Machine Learning, Support Vector Machine, Random Forest, Multimodal, PPMI.

1. INTRODUCTION

In 2019, almost 50 million people globally were afflicted with a neurodegenerative illness, frequently leading to dementia (a gradual deterioration of cognitive function caused by brain injury or disease); this figure is projected to increase to 152 million by 2060 (Armstrong, 2020; Allen et al., 2012).

The prevalence of neurodegenerative diseases resulting in dementia, according to the European dementia meta-analysis

(EURDEM) of research conducted in Europe, is 1.6% for men and 1% for women in the 65-69 age group, escalating to 11% for men and 12.6% for women in the 85-89 age group (Liu et al., 2014). Furthermore, Dorsey et al. (2006), research identify 3% of those over 65 affected by PD, this condition affects bodily muscular action. Though it becomes worse with time, it impacts posture, speech, and movement, causing tremors, muscular stiffness, and bradykinesia (Govindu and Palwe, 2023).

Early diagnosis and accurate projection of disease development are the foundations of effective therapy and improved patient outcomes, Myszczyńska et al. (2020); Noor et al. (2020). But they can be subjective, lengthy, and may delay necessary action until serious brain damage has occurred, clinical assessments are currently the main method used for diagnosis. The diverse range of PD symptoms and individual differences in treatment responses further make disease progression prediction difficult, Dadu et al. (2022); Severson et al. (2021). Therefore, by utilising multimodal data, such as neuroimaging, speech patterns, data from wearable sensors, and genetic information, recent ML developments provide encouraging options for enhancing early diagnosis and disease progression prediction, Soori et al., (2023); Bajwa et al., (2021). It is difficult to employ current ML models in clinical settings due to issues with interpretability,

2. LITERATURE REVIEW

Various researchers have employed distinct traits and datasets to forecast Parkinson's disease. This work is differentiated from existing research by creating diagnostic systems that utilise diverse data sources,

data heterogeneity, and low generalisability, Ciobanu-Caraus et al. (2024); Zhao et al. (2024). Furthermore, it is still a challenging process to integrate disparate data sources into a single framework, Chavali (2024).

This study bridges the gap between high-accuracy prediction and clinical interpretability by:

- 1. Integrating non-invasive multimodal data (olfactory, genetic markers, and demographics).
- 2. Developing a framework that specifically targets the **prodromal phase** to enable earlier intervention.
- 3. Providing feature-importance insights that align with clinical observations.

procedures, and tools capable of accurately analysing data to differentiate between individuals with Parkinson's disease and healthy individuals with high precision.

Table 1: Comparison of Methodologies and Research Gaps in Parkinson’s Disease Detection.

Research Category	Representative Studies	Limitations & Gaps	This Study’s Contribution
Single Modality (EEG/Neuro)	Loh et al. (2021); Aljalal et al. (2022); Yuvaraj et al. (2014, 2016); Sakar et al. (2013)	Small sample sizes (e.g., 31 subjects) and reliance on a single data source limit the ability to capture PD’s diverse symptoms.	Utilizes a large-scale, multimodal dataset (PPMI) to capture a wider spectrum of neurobiological and clinical signals.

Acoustic & Vocal Analysis	Chintalapudi et al. (2022); Little et al. (2009); Rana et al. (2022); Vaiciukynas et al. (2017)	While high accuracy is achieved (91.4%), these models do not account for non-motor symptoms like olfactory loss or genetic predisposition.	Integrates non-invasive clinical data, including olfactory (UPSIT) and genetic markers, to improve diagnostic robustness.
High-Cost / Invasive Multimodal	Zhang et al. (2023); Prashanth et al. (2016)	Often prioritizes cost-effectiveness over performance or relies on invasive/expensive biomarkers like CSF and SPECT imaging.	Prioritizes high-performance diagnostic precision (0.96 accuracy) using accessible, non-invasive clinical variables.
Deep Learning & Image Processing	Sivaranjini & Sujatha (2019); Shahid & Singh (2020)	Models like AlexNet achieve high accuracy but lack interpretability, which is essential for clinical validation and adoption.	Employs an interpretable Random Forest framework that provides feature importance to align with clinical observations.
Broad Clinical/ Lifestyle Surveys	Park et al. (2022); Cantürk & Karabiber (2016); Li et al (2017)	Attain lower accuracy (AUC 0.779) due to the inclusion of generalized variables that may lack specificity for early PD detection.	Specifically targets the Prodromal phase (3,026 cases) using specialized markers to enable early intervention.

Table 2: Comparative Analysis of Methodological Approaches and Research Gaps

Research Category	Representative Studies	Methodology & Key Findings	Limitations & Gaps	This Study's Contribution
Clustering & Dimensionality Reduction	Das (2010); Direito et al. (2017); Acharya et al. (2011)	Used Partial Least Squares (PLS) for reduction and Self-Organizing Maps (SOM) for clustering; identified diminished emotional expressiveness in PD patients.	Used Partial Least Squares (PLS) for reduction and Self-Organizing Maps (SOM) for clustering; identified diminished emotional expressiveness in PD patients.	Employs a supervised Random Forest framework optimized for high-accuracy (0.96) classification of prodromal stages.

Heuristic & Optimization Algorithms	Gupta et al. (2018); Karapinar Senturk (2020)	Employed an optimized Crow Search Algorithm and feature elimination techniques for early detection.	While accuracy improved, heuristic models can often lack the structural interpretability required for clinical validation.	Utilizes a transparent feature importance ranking that validates clinical markers like UPSIT scores alongside model performance.
Vocal Pattern Recognition	Rustempasic and Can (2013)	Utilized Fuzzy C-Means (FCM) clustering on vocalization data.	Achieved limited accuracy (68.04%) and low specificity (45.83%), suggesting vocal data alone is insufficient.	Integrates multimodal data streams (olfactory, genetic, and clinical) to achieve significantly higher diagnostic reliability.
Postural & Motor Biomarkers	Access (2014)	Classified patients based on postural instability using the L2 norm metric and SVM.	Postural instability is often a mid-to-late stage symptom, limiting its use in early-stage (prodromal) detection.	Focuses on the Prodromal phase using non-motor markers (smell and genetics) to identify the disease before motor symptoms peak.
Olfactory & Sleep Metrics	Silveira-Moriyama et al. (2008); Prashanth et al. (2014, 2016)	Used UPSIT, Sniffin' Sticks, and RBDSQ; later integrated CSF and SPECT markers to reach 96.40% accuracy.	High accuracy (96.40%) was reliant on expensive or invasive SPECT imaging and CSF measurements	Achieves comparable precision (0.96) using non-invasive clinical data, making the framework more accessible for routine screening.

Synthesis of Research Gap

The existing literature demonstrates that while machine learning can achieve high accuracy in PD detection, current approaches are often constrained by single-modality datasets, a lack of interpretability, or a reliance on invasive biomarkers. Studies focusing on single sources, such as EEG or vocal recordings, fail to fully affirm a diagnosis given the heterogeneity of symptoms and individual treatment responses. Furthermore, models achieving high accuracy through deep learning often function as "black boxes," hindering their clinical utility.

This study addresses these deficiencies by:

- Leveraging the PPMI Dataset: Utilizing a large-scale, multimodal, and non-invasive clinical dataset.
- Targeting the Prodromal Stage: Focusing on the earliest detectable phase of the disease to support timely clinical intervention.
- Optimizing Interpretability: Developing a Random Forest framework that combines rigorous data preprocessing (SMOTE and feature selection) with superior predictive performance, achieving a diagnostic accuracy of 0.96

3. PROPOSED Methodology

A flowchart of the recommended analysis shown in Figure 3.1, which demonstrates how a machine learning algorithm is constructed to detect indications of PD for early diagnosis. This research utilised data from PPMI, and for this investigation, many tables were amalgamated to enhance feature diagnosis. To accomplish the objectives of this study, data processing involved addressing outliers, consolidating tables, and eliminating missing values; a coefficient of variation was utilised to assess the relative dispersion of the data points and to equilibrate the dataset using the SMOTE method

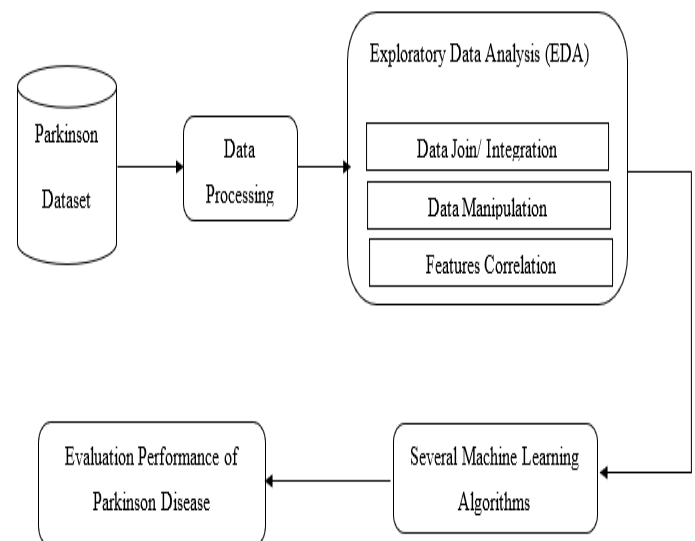


Figure 3.1: Methodology for the prompt diagnosis of the PD dataset.

3.1 Dataset Selection and Acquisition

PPMI is a multicentre observational study that gathers clinical and imaging data, together with biological samples from diverse cohorts, enabling researchers to identify signs of disease progression in Parkinson's disease. The official PPMI website (<https://ida.loni.usc.edu>) was accessed to acquire the latest information. This dataset resembles that utilised in Aljalal et al. (2022). The study utilized the PPMI longitudinal library (accessed March 2025), comprising 3,572 participants, categorized into three distinct cohorts: Prodromal (n = 2,766), PD (n =675), and Healthy Control (n =131).

3.2 Data Integration and Feature Engineering

The integration process involved consolidating disparate clinical tables into a unified analytical framework. Participant demographic information, family medical history, and personal health status were amalgamated to establish a comprehensive profile for each subject.

To optimize the model for early detection, the following specialized clinical indicators were incorporated:

- **Olfactory Function:** Data from the University of Pennsylvania Smell Identification Test (UPSIT) was integrated, as olfactory impairment is a recognized early-stage non-motor symptom of PD.
- **Neurological Imaging:** DATSCAN results were included to assess dopamine transporter levels in the brain, providing a physiological marker of neurodegeneration.
- **Clinical Symptomology:** Biological markers from blood tests and quantitative metrics of physical rigidity were added to represent common PD symptoms.

The final cleaned dataset contains 16 specific features across 3,572 entries. The detailed feature names and clinical descriptions are provided in Table 3.1.

Table 3.1: Feature name and its description

S/N	Feature Name	Description
1	PATNO	Participant ID
2	COHORT_x	Encoded Value for COHORT (1,2,4)
3	COHORT_DEFINITION	Decoded Value for COHORT ('Prodromal' 'PD' 'Healthy Control')
4	ENROLL_STATUS	Enrolment Status (Enrolled, withdrawn, and so on)
5	ENROLL_AGE	Age at Enrolment
6	ENRLPINK1	Pink1 Mutation at Enrolment
7	ENRLPRKN	Parkin Mutation at Enrolment
8	ENRLSRDC	Sporadic PD at Enrolment
9	ENRLHPSM	Hyposmia/ Generalized Risk at Enrolment
10	ENRLRBD	RBD at Enrolment
11	ENRLLRRK2	LRRK2 Mutation at Enrolment
12	ENRLSNCA	SNCA Mutation at Enrolment
13	ANYFAM_1	Any Family with PD (0: NO, 1: Yes)
14	TOTAL_CORRECT	Total Correct Items (0-40) smell identification score
15	UPSITFORM	The UPSIT form version administered
16	UPSIT_PRCNTGE	UPSIT Percentage 0 - 100

3.3 Exploratory Data Analysis (EDA)

3.3.1 Data Cleaning and Integration

During integration, the dataset was screened for completeness. Columns exhibiting a high degree of sparsity, defined as those containing more than 40% missing values, were excluded from the analysis to prevent bias and ensure the robustness of the statistical results.

3.3.2 Class labelling and Target Definition

For the purposes of early-stage classification, this study defines the diagnostic target as a binary outcome. Given that the Prodromal stage (Label 2) represents a high-probability precursor to clinical PD (Label 1), these two classes were merged into a single "PD-Positive" category. This resulted in a final target distribution of 3,441 PD-Positive cases and 131 Healthy Controls (Label 3). The relative proportions of these classes are visualized in Figure 3.2 and Figure 3.3.

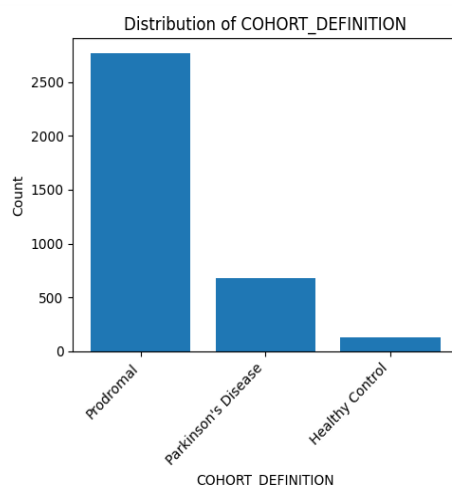


Figure 3.2: Proportion of PD outcome of the PPMI dataset.

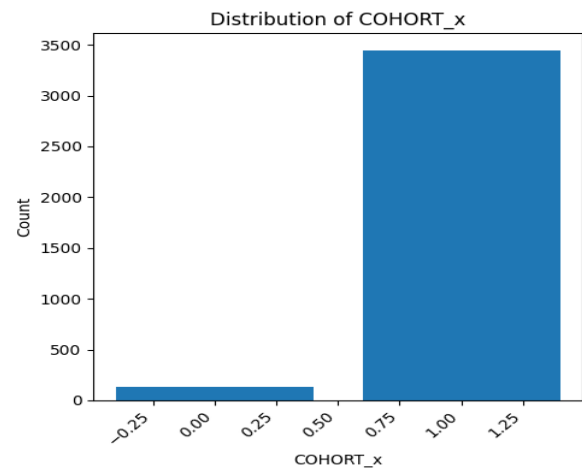


Figure 3.3: Proportion of Cohort_x outcome of the PPMI dataset.

3.3.3 Demographic Distribution

The age of the participants ranges from 30 to 89 years. Analysis of the age distribution (Figure 3.4) reveals a concentration of PD-positive outcomes in the **60 to 80-year** demographic, aligning with established epidemiological trends regarding the onset of neurodegenerative disorders.

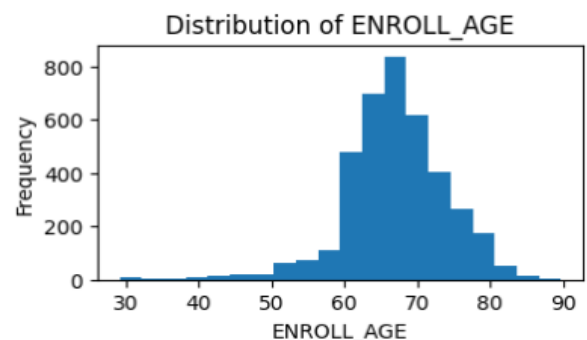


Figure 3.4: Histogram of the distribution of the age.

3.3.4 Other Features Distribution

This is the graphical representation of the other feature used in this study for the early prediction of PD.

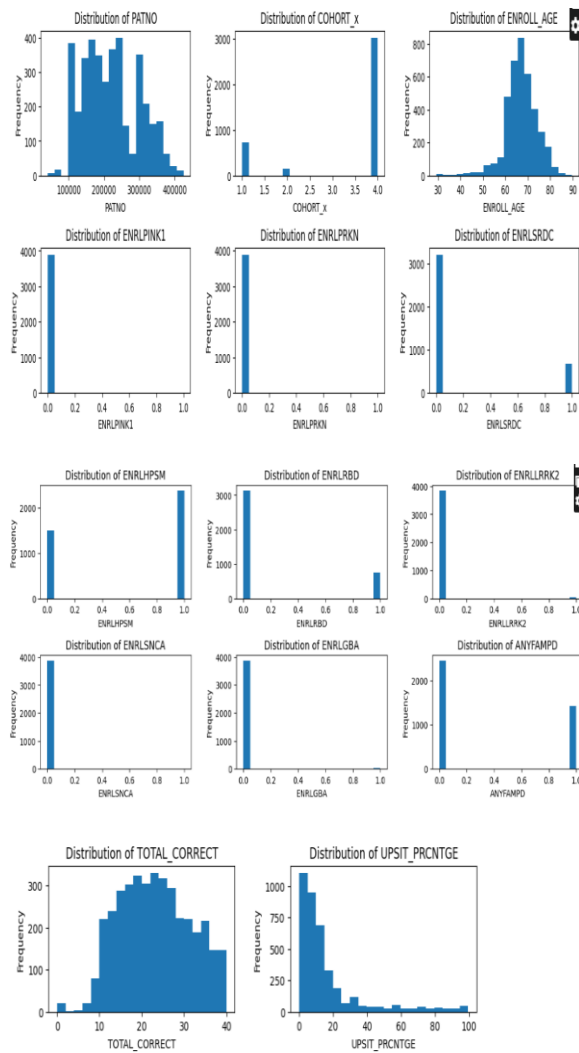


Figure 3.5 Other Attributes vs Outcome

3.4 Data Pre-processing

3.4.1 Handling Missing Data

Following the initial pruning of highly sparse features, a secondary validation confirmed that all remaining attributes were complete. Consequently, the dataset required no further imputation of missing values.

3.4.2 Feature Scaling (Normalization)

To standardize the scale of the diverse clinical features, ranging from age to

biochemical metrics, MinMaxScaler was utilized. This transformation maps all feature values to a normalized range between 0 and 1, ensuring that no single attribute disproportionately influences the model's objective function. The normalization was performed using the following formula:

$$X_{min} = (X - X_{min}) / (X_{max} - X_{min})$$

3.4.3 Correlation Analysis

Pearson's correlation coefficients were computed to evaluate the linear relationships between the 16 features. This step is critical for identifying potential multicollinearity and ensuring that the selected clinical markers provide distinct diagnostic information. The resulting correlation matrix is presented in Figure 3.6.

$$\rho_{X,Y} = \text{covariance}(X, Y) / (\text{Std } X \cdot \text{Std } Y)$$

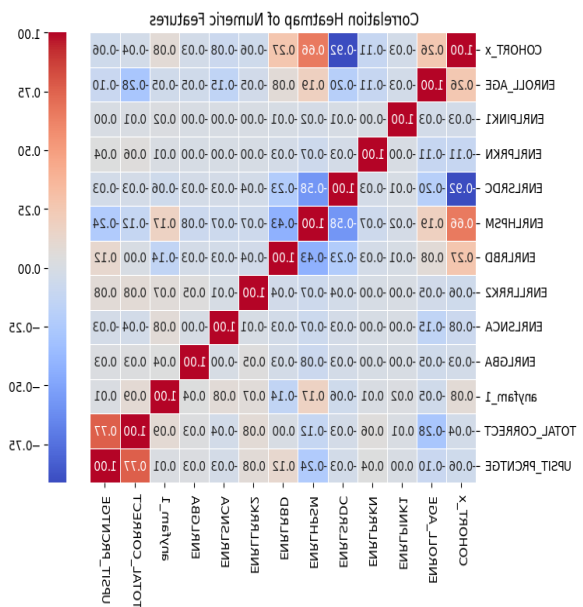


Figure 3.6 Correlation between features.

4. EXPERIMENTATION

4.1 Machine Learning approach towards PPMI Data

This chapter details the application of supervised classification to the Parkinson's Progression Markers Initiative (PPMI) dataset. The feature set consists of sixteen (16) attributes, including two categorical variables; the 'cohort_x' column serves as the target output, while the remaining attributes function as predictive inputs.

The experimental workflow was organized into four distinct phases:

1. Initialization of experiments using Support Vector Machine (SVM) and Random Forest (RF) algorithms.
2. Execution of baseline models using a 70/30 training-validation split.
3. Implementation of the Synthetic Minority Over-sampling Technique (SMOTE) to resolve class imbalance issues.
4. Comparative analysis of the results against pertinent existing research.

4.2 Algorithm Selection and Justification

The study employed SVM and RF due to their established efficacy in medical diagnostic modeling.

- Random Forest (RF): Selected for its inherent robustness to noise, capability to model non-linear relationships, and built-in feature importance estimation, which enhances clinical interpretability.
- Support Vector Machine (SVM): Utilized as a high-performance

baseline due to its effectiveness in handling high-dimensional clinical spaces.

Performance was evaluated using standard clinical metrics: accuracy, precision, and recall.

4.3 Implementation and Refinement of Random Forest

Initial RF implementation on the raw 70/30 split resulted in an accuracy of 100%. While this indicated a high fit, it also signaled significant overfitting driven by class imbalance. To achieve a more reliable model, SMOTE was applied for dataset balancing, and highly correlated features were removed based on a Pearson correlation analysis.

Following decorrelation and resampling, the RF model achieved a stable and clinically relevant performance of 96% accuracy. The final metrics are detailed in Table 4.2, with corresponding ROC curves and confusion matrices provided in Figures 4.1 and 4.2.

4.4 Implementation and Refinement of Support Vector Machine

Similarly, the initial SVM baseline reached 100% accuracy, an overly optimistic result suggesting model overfitting. After refining the feature set and applying SMOTE-based oversampling, the SVM achieved a stabilized performance with 93% accuracy. While robust, the SVM results remained secondary to the performance of the Random Forest model. Refined metrics are summarized in Table 4.4, with the confusion matrix and ROC curve illustrated in Figures 4.3 and 4.4.

MODEL	Normal				<u>Undersampling/Oversampling</u>			
	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)
RF	100	99	98	97	100	100	100	100

Table 4.1 Random Forest result on the dataset

MODEL	Normal				Over Sampling			
	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)
RF	97	76	73	71	96	96	96	96

Table 4.1 Random Forest result after a highly correlated feature was removed.

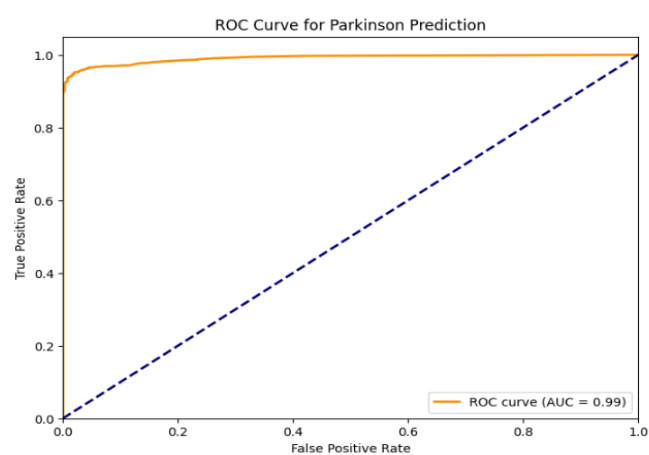


Figure 4.1: ROC FOR OVERSAMPLING

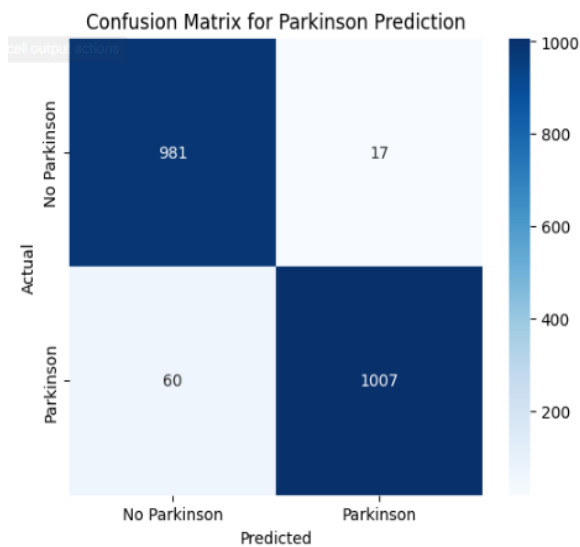


Figure 4.2 Confusion matrix for Oversampling data

True Negative (TN) = 981, False Positive (FP) = 17

True Negative (TN) = 60, True Positive (TP) = 1007

$$\text{Accuracy} = \frac{TP + TN}{FP + FN + TP + TN} \quad (4.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

Where:

No Parkinson -> Negative, Parkinson -> Positive.

MODEL	Normal				Under Sampling			
	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)
SVM	100	99	100	99	100	100	100	100

Table 4.3 SVM result for dataset.

MODEL	Normal				Over Sampling			
	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)	Accuracy (%)	Precision (%)	F1-Score (%)	Recall (%)
SVM	96	74	67	64	93	93	93	93

Table 4.4 SVM result for the dataset after removal of the highly correlated features

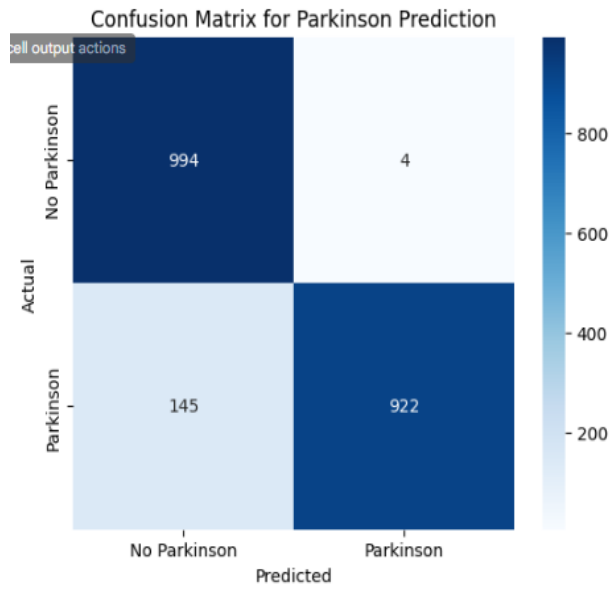


Figure 4.3: Confusion matrix results for SVM

True Negative (TN) = 994, False Positive (FP) = 4

True Negative (TN) = 145, True Positive (TP) = 922

$$Accuracy = \frac{TP + TN}{FP + FN + TP + TN} \quad (4.1)$$

$$Precision = \frac{TN}{TN + FP} \quad (4.2)$$

$$Recall = \frac{TP}{TP + FN} \quad (4.3)$$

Where:

No Parkinson = Negative, Parkinson = Positive.

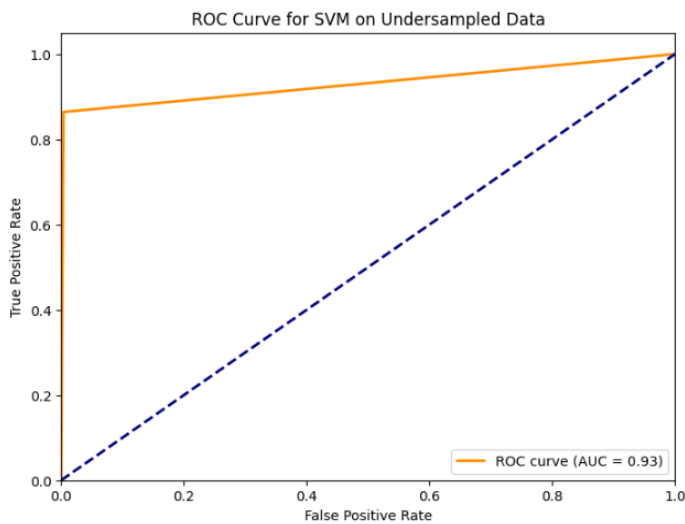


Figure 4.4: ROC curve for SVM

4.6 Comparison with Existing Literature

A comparative analysis was conducted between the proposed RF framework and existing models in the literature. As shown in **Table 4.5**, this study's accuracy of **96%** surpasses several benchmarks, including the **GenoML** framework (85.03%) and **XGBoost** models (94.0%).

Table 4.5: Performance Comparison Against Prior Research

Authors (Year)	Models	Accuracy (%)
This Study	SVM	93%
	RF	96%
Park et al., (2022)	RF	77.9%
Makarious et al., (2022)	GenoML	85.03%
Zhang et al., (2023)	XGBOOST	94%
(Govindu and Palwe, 2023)	RF	91.83%
Sivaranjini and Sujatha, (2019)	AlexNet	88.9%

4.7 Summary

The experimental results demonstrate that the proposed multimodal framework effectively classifies PPMI data for early PD detection. The Random Forest algorithm proved to be the superior model, achieving **96% accuracy** while offering greater

robustness and interpretability compared to the SVM baseline. By relying on accessible, non-invasive clinical data, this framework provides a significant advancement over existing diagnostic methods.

5. CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

Parkinson's Disease (PD) remains a complex neurological condition with an inadequately elucidated etiology and a lack of curative therapies. Early identification, particularly during the **prodromal phase**, is critical for facilitating timely clinical intervention and improving long-term patient outcomes. This research successfully developed a cost-effective, non-invasive framework for predicting PD risk by leveraging multimodal data from the **PPMI dataset**.

The study methodically evaluated the predictive efficacy of **Random Forest (RF)** and **Support Vector Machine (SVM)**. Our findings demonstrate that models utilizing demographic variables, clinical evaluations, and genetic data can achieve predictive performance comparable to—and in some cases exceeding—models that incorporate invasive biomarkers. Specifically, the Random Forest model emerged as the superior technique, achieving a diagnostic accuracy of **96%**. The model's ability to capture complex feature interactions while remaining robust after feature refinement underscores its suitability as a clinical decision support tool.

A primary outcome of this research is the identification of **olfactory function (UPSIT)** and personal health records as the most significant predictors of PD risk. We assert that this non-invasive approach provides a feasible pathway for early diagnosis,

reducing the clinical reliance on expensive or invasive diagnostic procedures.

5.2 Constraints and Limitations

While this research yields valuable diagnostic insights, several limitations must be acknowledged:

- **Dataset Scope:** The analysis relies exclusively on the PPMI dataset; consequently, external validation using independent, diverse cohorts was not performed in this stage of the research.
- **Data Integrity:** Inconsistencies and missing information within the original data dictionary posed significant challenges during the preprocessing phase.
- **Clinical Readiness:** Extensive clinical validation in real-world settings is a necessity before this framework can be deployed for frontline medical use.

5.3 Recommendations for Future Work

To enhance the generalizability of these findings, future research should focus on:

1. **Dataset Diversity:** Utilizing larger, globally diverse datasets to ensure model robustness across various ethnic and geographic populations.
2. **Algorithmic Expansion:** Incorporating and evaluating additional machine learning architectures, such as Gradient Boosting Machines or Ensemble methods, to further refine accuracy.
3. **Interdisciplinary Collaboration:** Establishing closer partnerships between medical professionals and

data scientists to translate these computational advancements into actionable bedside technology.

5.4 Contribution to Knowledge

This research contributes to the field of computational neuroscience and medical informatics in the following ways:

1 Model Optimization:

Demonstrating that classical machine learning models, specifically Random Forest, can outperform more complex deep learning approaches when paired with rigorous feature engineering and evaluation.

2 Diagnostic Framework:

Establishing a non-invasive, cost-effective framework for PD risk prediction that prioritizes patient accessibility.

3 Feature Identification:

Identifying and validating key predictive clinical features that directly support early-stage diagnosis and evidence-based clinical decision-making

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